Proceedings of the International Conference on Artificial Intelligence & Engineering · Published 2021

Hierarchical Planning and Shared Semantic Memory for Autonomous Multi-Agent Systems

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ABSTRACT

Autonomous multi-agent systems often experience task desynchronization and communication bottlenecks in complex environments. We propose a hierarchical planning framework coupled with a shared semantic memory graph that coordinates decentralized agents. Our architecture decomposes high-level objectives into localized sub-plans while synchronizing state representations through graph-based embeddings. Empirical evaluations across cooperative multi-agent benchmarks demonstrate a 34% reduction in planning latency and a 28% improvement in task completion rate. However, the framework suffers from memory synchronization overhead as the number of active agents scales beyond fifty, leaving real-time scalability an open challenge.

Keywords— Artificial Intelligence, Multi-Agent Systems, Hierarchical Planning, Semantic Memory

INTRODUCTION

Multi-agent coordination in dynamic domains requires balancing decentralized autonomy with global coherence. Prior approaches rely on centralized controllers that present single points of failure, or fully decentralized protocols that suffer from state divergence. In this paper, we bridge this gap through hierarchical task networks supported by shared memory graphs.

METHODOLOGY

Our system establishes a two-tiered hierarchy: high-level strategic planners that broadcast symbolic sub-goals, and low-level tactical agents that execute reactive policies. The shared memory graph is updated using lightweight delta broadcasts, maintaining consistency through vector clocks.

RESULTS

We evaluated our system in simulated disaster response and robotic warehousing environments against baseline centralized and peer-to-peer protocols. The hierarchical planning mechanism achieved a 28% higher task success rate while reducing redundant message exchanges by 41%.

LIMITATIONS

The primary limitation is the communication overhead during network partitions. When network latency spikes, semantic memory synchronization can introduce stale state reads, occasionally causing redundant sub-task allocations.

REFERENCES

[1] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[2] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[3] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.